PRACTICAL ASSESSMENT

Recycling Plant Project

PLC control of a bidirectional conveyor with counting and container monitoring

Assessment information	Details
Module	Programmable Logic Controllers 226 PLC216D
Assessment mode	Laboratory demonstration and project report
Assessment type	Group practical assessment; 5 Members per group
Controller	Siemens SIMATIC S7-1200 CPU 1211C DC/DC/DC
Available onboard I/O	6 digital inputs, 2 analogue inputs and 4 transistor outputs
Additional features	Object counting, full-container monitoring and transport timeout detection
Submission format	Report using the prescribed headings and a TIA Portal project
Due date	25-10-2026
Total marks	60 marks proposed; subject to lecturer approval



1 Assignment overview

A recycling plant needs to separate copper pieces from plastic pieces. A metal detector identifies metallic objects, and the conveyor carries each object to the appropriate collection container by changing direction. Red and green indicators show the classification. The improved system also confirms delivery, counts the sorted objects and stops when a container is full or transport fails.

For the laboratory demonstration, feed separated, detectable pieces one at a time at the central sorting point. Do not feed a continuous mixture of ground material: reversing a belt would move the entire mixture rather than separate its individual components. The model classifies the supplied non-metallic samples as plastic; it does not identify all types of non-metallic material.

This assessment develops the original recycling plant practical into a monitored sorting system. The conveyor motor remains part of the output system, controlled through a bidirectional driver. The three additional features must fit the onboard I/O allocation without an expansion module.

2 Learning outcomes

Design, wire, program and demonstrate a PLC sorting system that retains the original sorting functions while adding monitoring within the available onboard I/O.

- Identify suitable digital and analogue sensors and prepare an allocation table.
- Implement a latched operating command, classification memory and a controlled sorting sequence.
- Drive a DC conveyor motor in both directions through a suitable motor driver.
- Use counters, timers and analogue comparisons to monitor production and faults.
- Demonstrate interlocking, controlled reset and safe restart behavior.

3 Task

Design, construct, program and test a laboratory system that sorts one metallic or plastic piece at a time. Retain the original START, emergency-stop, bidirectional motor and red/green indicator functions. Complete the following tasks and support component choices with datasheets or other credible technical sources.

1. Select the motor, compatible driver, metal detector, object sensors and two 0–10 V level sensors.
2. Prepare the power wiring diagram and verify all voltage and current requirements.
3. Confirm the physical I/O addresses in TIA Portal and create symbolic tags.
4. Develop a sequential function chart and ladder program for the sorting cycle.
5. Implement separate metal and plastic counts based on confirmed delivery.
6. Scale the two-level signals and implement full-container thresholds with hysteresis.
7. Implement a transport timer, latched fault indication and controlled reset.
8. Demonstrate every verification test and submit the report and PLC project.

4 Hardware and input output guide

Component	Required use
CPU 1211C DC/DC/DC	6 digital inputs, 2 analogue voltage inputs and 4 transistor outputs.
24 V DC supply	Power the PLC and compatible field devices. Size supply and protection for the actual loads.
DC motor and conveyor	Carry one object from the central sorting point to either end.
Bidirectional motor driver	Accept PLC-compatible commands and supply the motor current.
Inductive sensor	Detect the copper or other metallic test pieces at the sorting point.
Three object sensors	One presence sensor at the sorting point and one collection sensor at each end.
Two analogue level sensors	Provide 0–10 V signals representing container fill level.
START and emergency stop	Provide start command and independent emergency-stop action with PLC feedback.
Red and green indicators	Display classification; use flashing red for an inhibited or fault condition.
Engineering station or optional HMI	Show counts, levels and fault status; issue permitted reset commands.

Power and interface design

Use a 24 V DC distribution supply for the PLC and compatible sensors. Where the original breadboard circuit needs 5 V DC, use a regulated step-down converter from that supply. Check each input and output interface before connection. A 5 V circuit must not be exposed directly to a 24 V PLC output.

Use sensors whose electrical outputs suit the PLC input wiring. Source field-device power from the protected supply distribution and comply with the PLC sensor-supply current limits. The motor power circuit must have its own appropriately protected branch; it must not draw motor current through a PLC signal output or the PLC sensor-supply terminals.

Motor and emergency stop

Q0.0 and Q0.1 command the driver; the motor connects to the driver power outputs. Use a driver or interface designed for 24 V PLC signals, the motor voltage, starting current and stall current. Implement both software direction interlocking and a suitable hardware arrangement that prevents conflicting drive commands.

The emergency-stop device must disable the motor power or drive through an appropriate hardware safety circuit. I0.1 is feedback to the standard PLC, not the safety function itself. Releasing the emergency stop must not restart the conveyor. Require a fresh START command after the system is ready.

The allocation below uses the CPU onboard I/O without an expansion module. Addresses are proposed and must be confirmed in the TIA Portal device configuration before wiring or programming.

Address	Tag	Device and function
I0.0	StartPB	Normally open START button; rising edge requests operation.
I0.1	SafetyOK	Emergency-stop circuit feedback; 1 means the circuit permits operation.
I0.2	MetalDetected	Inductive sensor at the classification point.
I0.3	ObjectPresent	Presence sensor at the central sorting point.
I0.4	MetalExit	Confirms arrival at the metal container.
I0.5	PlasticExit	Confirms arrival at the plastic container.
IW64	MetalLevelRaw	0–10 V analogue level sensor for the metal container.
IW66	PlasticLevelRaw	0–10 V analogue level sensor for the plastic container.

Address	Tag	Output device and function
Q0.0	MotorForward	Driver forward command; carries plastic towards its container.
Q0.1	MotorReverse	Driver reverse command; carries metal towards its container.
Q0.2	RedLED	Metal classification; flashing red overrides this indication during a fault or full-container stop.
Q0.3	GreenLED	Plastic classification during a normal sorting cycle.

I/O usage: 6 of 6 digital inputs, 2 of 2 analogue inputs and 4 of 4 digital outputs. There is no spare physical I/O for a dedicated reset button, buzzer or additional full-container lamp.

Meaning of IW64

I means input, W means word, and 64 is the starting byte address. An input word contains 16 bits and occupies two bytes, so IW64 uses bytes 64 and 65. IW66 uses bytes 66 and 67. An analogue channel provides a numerical reading; a digital input such as I0.0 supplies a Boolean ON or OFF state. Use symbolic tags in the program and verify the actual channel addresses assigned by TIA Portal.

5 Additional feature requirements

Automatic counting of sorted objects

Maintain separate MetalCount and PlasticCount values. Increment the relevant counter only when the expected collection sensor changes from OFF to ON during its active transport state. Do not count merely because the classification sensor detects an object. These records confirmed delivery rather than attempted sorting.

Use edge detection and a cycle-complete latch to prevent a long sensor signal from producing multiple counts. Require the collection sensors to clear before accepting a new cycle. Make counter reset available through an

HMI or a supervised TIA Portal watch-table command, only while the conveyor is stopped and no object is being transported. Resetting counts must not reset a transport fault or start the motor.

Full container detection using analogue inputs

Scale both 0–10 V level signals to a calibrated fill percentage. Configure each sensor so that its output increases with fill level or invert the scaling if the selected distance sensor gives the opposite relationship. Compare the scaled value with a full threshold for each container.

For an illustrative classroom configuration, declare FULL at 90% and clear it only below 80%. These values are proposed settings and must be calibrated for the actual container and sensor. Apply suitable filtering to reduce false indications caused by moving material. Stop operation when either container is full, show the affected container on the HMI or watch table, and flash the red indicator.

After emptying, the full condition clears when the reading falls below its clearing threshold. The run latch remains OFF until START is pressed. A basic 0–10 V reading may not reliably distinguish an empty container from a disconnected sensor; sensor diagnostics require a suitable device or additional measures.

Transport timeout and fault detection

Start a TON transport timer when the forward or reverse motor command is issued. Stop and reset the timer after the expected exit sensor confirms delivery. If the object does not arrive before the preset expires, turn both motor commands OFF, latch TransportFault and flash the red indicator.

Determine the timeout from repeated normal travel measurements, then allow for expected variation. A 5 s setting may be used as an initial classroom example only after confirming that normal travel is shorter. This feature identifies failure to deliver; it cannot distinguish a mechanical jam from a failed motor, slipping belt or faulty sensor.

After the obstruction or equipment fault is cleared, remove the affected object and return the sorting point and collection sensors to a clear condition. Accept FAULT RESET only when these conditions are satisfied, the safety circuit is healthy and both containers are below full. Reset returns the system to stopped idle; a separate START command is needed.

6 Important implementation notes

State	Actions	Condition for next state
Stopped	Both motor commands OFF. Ignore object arrivals.	Fresh START, healthy safety feedback, no fault, no full container and sensors clear.
Ready	Wait with the motor OFF for one object at the central sensor.	ObjectPresent becomes ON.
Classify	Allow the sensor to settle; latch metal or plastic classification.	Classification interval completes while presence is confirmed.
Direction delay	Keep both motor commands OFF for the selected delay.	Delay is completed and all operating permissions remain healthy.

State	Actions	Condition for next state
Transport metal	MotorReverse ON; red LED ON; run transport timer.	MetalExit rising edge confirms delivery.
Transport plastic	MotorForward ON; green LED ON; run transport timer.	PlasticExit rising edge confirms delivery.
Complete	Motor OFF; increment the correct counter once.	Central and both collection sensors clear; return to Ready.
Inhibited or fault	Both motor commands OFF; red LED flashes.	Clear full condition or perform valid fault reset, then press START.

Safety feedback, container-full flags and fault flags override every operating state. Any loss of permission clears the run latch and removes motor commands. Power restoration also leaves the system stopped. If a stop interrupts transport, clear the object and sensors before starting a new cycle; do not assume its position is known.

Classification and direction control

Only interpret absence of metal detection as plastic when ObjectPresent confirms an object. Position both sensors so they examine the same piece. Latch the classification before moving it, because the inductive sensor may turn OFF as the piece leaves. Choose the classification settling interval through laboratory testing.

Never energise MotorForward and MotorReverse together. Each command requires the opposite command to be OFF. Use a stopped interval appropriate to the driver and conveyor before a change of direction. A proposed 0.5 s delay must be checked against the actual motor and driver behavior.

Indicator priority

During normal transport, show red for metal and green for plastic. When full or a transport fault is active, switch green OFF and flash red. The HMI or watch table identifies the specific cause. No additional buzzer or container indicator is included in this four-output allocation.

Use a state-based program in TIA Portal with symbolic tags. Separate input processing, analogue scaling, state transitions, counting, fault handling and final output assignment. Assign each physical output in one place so that the stop conditions have clear priority.

Program element	Purpose
RunEnabled and state variable	Store the operator run request and the current sorting state.
LatchedMetal	Remember the classification until the cycle is completed or is aborted.
MetalCount and PlasticCount	Store confirmed delivery totals using counters or integer values.
MetalFull and PlasticFull	Store hysteresis-based full conditions.
TransportFault	Latch an unsuccessful delivery until a valid reset.

Program element	Purpose
Classification and direction timers	Allow reliable classification and stopped direction changes.
Transport TON	Detect delivery that exceeds the configured travel time.
ResetFaultCmd and ResetCountsCmd	Momentary HMI or supervised test commands; not physical inputs.

Analogue scaling example

For the standard Siemens unipolar voltage representation, verify that the selected channel maps 0–10 V to 0–27648 raw units. With a linear, calibrated sensor where 0 V means empty and 10 V means full, calculate FillPercent = RawValue × 100.0 / 27648.0. Use real-number arithmetic. For other sensor mappings, scale from the measured empty and full calibration values instead.

At an illustrative 90% full threshold, the corresponding raw value is approximately 24883; the 80% clearing threshold is approximately 22118. These values apply only to the stated linear mapping. Compare scaled values after calibration and document how invalid or out-of-range readings are handled.

Operator display and reset access

An HMI may communicate with the PLC without consuming onboard digital I/O. Display system state, classification, both counts, both fill percentages and fault flags. Provide separate counter-reset and fault-reset commands with the stated permissions. If no HMI is available, use a supervised watch table for assessment and return momentary command tags to FALSE after use. Do not force motor outputs to bypass program interlocks.

Commissioning settings to record

Record the actual CPU order number, firmware, TIA Portal version, input addresses, sensor models, supply ratings, driver model, conveyor travel times, timer presets, full and clearing thresholds, analogue calibration points and counter retention policy. Specify whether counts reset or persist after power loss; verify the configured behavior.

The specified I/O has no spare digital input for a separate normal STOP button. The original emergency-stop function is retained. If an operational STOP command is needed, implement it through the optional HMI and clear Run Enabled immediately. It does not replace the hardware emergency stop.

7 Required report structure

Organize your submission using the following headings. Include labelled diagrams and source citations where they help explain the design.

1. Cover page: title, student names, student numbers, module, lecturer and submission date.
2. Introduction: recycling scenario, laboratory scope, aim and three improvements.
3. Hardware selection and wiring: component models, electrical compatibility, power distribution and emergency-stop arrangement.
4. Input and output allocation: confirmed addresses, symbolic tags, functions and explanation of analogue words.

5. Control design: sequential function chart, ladder program, timers, counters, interlocks, state transitions and reset rules.
6. Additional feature investigations: complete the template below for each improvement.
7. Verification results: test procedure, observed result, pass or fail and supporting evidence.
8. Conclusion: evaluate sorting performance, limitations and possible improvements.
9. References: use one consistent academic referencing style.

8 Feature investigation template

Repeat this structure for automatic counting, full-container monitoring and transport timeout detection. Replace the prompts with your design decisions and observed evidence.

Required field	Student response
Feature name and purpose	State what the feature adds to the original system.
Associated inputs and outputs	List confirmed addresses and symbolic tags.
Hardware or data used	Identify sensor/driver models or PLC variables.
Operating principle	Explain the signal path and how the PLC responds.
Program elements	Identify counters, timers, state flags and edge detection.
Settings and calibration	Record thresholds, delay times and calibration method.
Reset and restart conditions	State when reset is accepted and whether START is required.
Verification evidence	Record test inputs, expected and observed behavior.
Limitations and sources	Explain constraints and cite technical sources.

9 Summary and verification table

Feature	Signals used	PLC function	Expected response
Counting	I0.4 and I0.5	Counters and edge detection	One count per confirmed delivery.
Container monitoring	IW64 and IW66 proposed	Scaling and hysteresis	Stop when either container is full.
Transport timeout	Active motor command and expected exit sensor	TON and latched fault	Stop and flash red if delivery exceeds the preset.

Test	Expected result
Metal sample	Reverse transport, red LED and one metal count after arrival.
Plastic sample	Forward transport, green LED and one plastic count after arrival.

Test	Expected result
No object present	No plastic classification or motor movement.
Collection sensor stays ON	One count only; new cycle blocked until sensors clear.
Either container reaches FULL	Both directions commands OFF; red flashes; affected container identified.
Container emptied	No automatic restart; START is required.
Block or delay delivery	Timeout latches faults and stops both direction commands.
Fault reset before clearance	Reset rejected; motor remains OFF.
Valid fault reset	Fault clears; system remains stopped until START.
Emergency stop during either direction	Hardware drive disabled and run latch cleared; release does not restart.
Direction change	No overlapping commands; stopped delays occurs.
Counter reset and power cycle	Reset permissions and documented retention behavior are correct.

Add columns for observed results and pass or fail to your submitted test record. Record the actual settings used and include evidence of the output state, counts and fault flags.

10 Assessment rubric

This proposed rubric retains 40 marks for the original functions and adds 20 marks for the three improvements. It requires lecturer approval before use. Award marks within each row according to the evidence demonstrated.

Criterion	Excellent	Competent	Developing or incomplete	Marks
Hardware and power	Correct wiring, compatible devices and protected supply.	Works with minor documentation omissions.	Incorrect or incomplete wiring or interfaces.	14
Motor driver	Compatible driver and verified interlocking.	Direction control works with minor explanation gaps.	Driver or direction control incomplete.	4
Indicators	Correct metal and plastic indication.	Mostly correct with minor inconsistencies.	Missing or incorrect indication.	6
Bidirectional sorting	Both materials delivered to the correct containers.	Sorting works with occasional sequence errors.	One direction fails or sorting unreliable.	10
START and emergency stop	Correct start and independent stop; no automatic restart.	Functions work with minor evidence gaps.	Start/stop incomplete or restart uncontrolled.	6

Criterion	Excellent	Competent	Developing or incomplete	Marks
Counting	Confirmed delivery counted once; reset controlled.	Counts generally correct; minor reset/edge issues.	Counts missing, duplicated or based only on detection.	6
Container monitoring	Calibrated scaling, full response and hysteresis verified.	Full response works; limited calibration evidence.	Scaling or full response missing or incorrect.	7
Transport fault	Timeout, latch, indication and controlled reset verified.	Fault response works; some recovery evidence missing.	Fault fails to stop transport or reset is uncontrolled.	7
TOTAL				60

11 Submission checklist

☐ I included the project scope and the three additional features.

☐ I used no more than six digital inputs, two analogue inputs and four digital outputs.

☐ I included the conveyor motor driver and both directional commands in the output allocation.

☐ I confirmed addresses in TIA Portal and explained IW64 and IW66.

☐ I included component models, datasheets and labelled wiring diagrams.

☐ I included the sequential function chart and commented on ladder program.

☐ I demonstrated counting, analogue calibration, full-container response and transport timeout.

☐ I verified emergency stop, direction interlocking and restart behaviour.

☐ I completed the feature investigation templates and recorded test results.

☐ I included a conclusion, consistent references and the TIA Portal project.

12 Academic integrity

The submitted work must be your own. Paraphrase and acknowledge information from sources. Cite borrowed diagrams, code examples, images, definitions and device specifications. Report actual observed test results and clearly distinguish simulation results from hardware demonstrations. Follow the institution's assessment policy and the lecturer's instructions on permitted tools and collaboration.

Reference documents

1. Siemens CPU 1211C DC/DC/DC data sheet, order number 6ES7211-1AE40-0XB0. Onboard I/O and power specifications. https://support.industry.siemens.com/teddatasheet/?caller=SIOS&format=pdf&language=en&mlfbs=6ES7211-1AE40-0XB0
2. Siemens S7-1200 Programmable Controller System Manual, V4.7, December 2024. Confirm analogue representation, wiring and device-specific limits against the installed CPU. https://cache.industry.siemens.com/dl/files/302/109977302/att_1307432/v1/s71200_system_manual_en-US.pdf